# A DENSE CONSTRUCTIBLE SUBSET CONTAINS A DENSE OPEN SUBSET

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The statement that a finite union of nowhere dense subsets is nowhere dense. Invoked inside the counted proof (source line 4325). Combines the closures of the locally closed pieces of the complement.

*Source reference.* Original lemma lemma-nowhere-dense: sources/stacks-topology/pinned-original-source.tex, source lines 4222–4222. Navigation pointer only: the counted text refers to the Stacks item lemma-nowhere-dense, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

**Lemma 0.1.** *Let $X$ be a topological space. Let $E \subset X$ be a finite union of locally closed subsets (e.g. $E$ is constructible). Then $E$ contains a dense open subset of its closure.*

*Proof.* After replacing $X$ by the closure of $E$ we may assume that $E$ is dense in $X$ and we have to show that $E$ contains a dense open. Then the complement $T$ of $E$ is a finite union of locally closed subsets, say $T = T_1 \cup \ldots \cup T_n$, and $T$ is nowhere dense in $X$. Since $T_i$ is open dense in its closure $Z_i$ we see that also $Z_i$ is nowhere dense in $X$. Then the closure $Z = Z_1 \cup \ldots \cup Z_n$ of $T$ is nowhere dense in $X$ by Lemma lemma-nowhere-dense. We conclude that $E$ contains the dense open $X \setminus Z$. □

*Date*: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.